

FIITJEE

ALL INDIA TEST SERIES

OPEN TEST

JEE (Main)-2024

TEST DATE: 14-01-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. C
Sol. Conservation of mechanical energy with respect to point P

$$\frac{1}{2}(4)(2+1)^2 = \frac{1}{2}(100)(\Delta\ell)^2$$

where $\Delta\ell$ is maximum elongation in the spring

So, $\Delta\ell = 6$ cm

$$\text{So, maximum acceleration} = \frac{(100)(6)}{4} = 150 \text{ m/s}^2$$

2. C
Sol. The block will start slipping when

$$0.3 \times 20 = \frac{8}{4}t \Rightarrow t = 3 \text{ sec}$$

From impulse momentum theorem

$$v = \frac{\frac{1}{2}(8+6) + 8 - 0.25 \times 20 \times 2}{2} = 2.5 \text{ m/s}$$

3. D

$$\text{Sol. } U = (1)\left(\frac{5}{2}R\right)T + (1)\left(\frac{3}{2}R\right)T = 4RT$$

$$\frac{v_{r/m}}{v_{r/H}} = \sqrt{\frac{2}{3}}$$

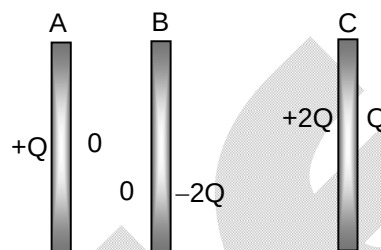
$$\frac{v_{s/m}}{v_{s/He}} = \sqrt{\frac{6}{5}}$$

$$\frac{v_{r/He}}{v_{r/H}} = \sqrt{\frac{2}{4}} = \frac{1}{\sqrt{2}}$$

4. C

Sol. At steady state charges on each surfaces are

$$\text{So, } V_C - V_A = \frac{2Q(2d)}{\epsilon_0 A} = \frac{4Qd}{\epsilon_0 A}$$

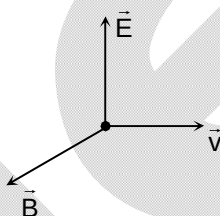


5. B

 Sol. For equilibrium $\vec{E} = \vec{B} \times \vec{v}$

$$\text{So, } \vec{E} \times \vec{B} = (\vec{B} \times \vec{v}) \times \vec{B}$$

$$\Rightarrow \vec{E} \times \vec{B} = |\vec{B}|^2 \vec{v}$$



6. A

$$\text{Sol. } \vec{v}_{\text{image/lens}} = \left(\frac{v}{u}\right)^2 \vec{v}_{\text{object/lens}}$$

$$\frac{1}{v} = \frac{1}{20} - \frac{1}{30} \Rightarrow v = +60 \text{ cm}$$

$$\vec{v}_{\text{image/lens}} = (2)^2(4\hat{i}) - 2\hat{i} = 14\hat{i} \text{ m/s}$$

 $\hat{i}$ unit vector is towards right

7. D

 Sol. There is no change in wavelength of characteristic spectrum due to change in energy of the incident electrons. K_α wavelength is large as compared to K_β wavelength.

8. B

$$\text{Sol. } \vec{a} = -\frac{\pi^2}{64} \vec{x}, \text{ so } \omega = \frac{\pi}{8} \text{ rad/s}$$

$$A \frac{\pi}{8} = 4\sqrt{2} \text{ (because the particle is having time period of 16 sec)}$$

$$\text{So, } A = \frac{32\sqrt{2}}{\pi} \text{ m}$$

9. D

$$\text{Sol. } i = \frac{10 - 0.7}{1.2} = 7.75 \text{ mA}$$

10. B

 Sol. Due to temperature increment all d_1 , d_2 and r increases.

11.

C

Sol.

$$W_{AB} = W_{CD} = 0$$

$$W_{BC} = nR(500 - 350) = 150nR$$

$$W_{DA} = nR(300 - 400) = -100nR$$

$$\Delta Q_{AB} = n \frac{3}{2} R(350 - 300) = 75nR$$

$$\Delta Q_{BC} = n \frac{5}{2} R(150) = 375nR$$

$$\Delta Q_{CD} = n \frac{3}{2} R(-100) = -150nR$$

$$\Delta Q_{DA} = n \frac{5}{2} R(-100) = -250nR$$

$$\text{So, } \eta = -\frac{W_{\text{net}}}{\Delta Q_{\text{input}}} = \frac{50nR}{450nR} = \frac{1}{9}$$

12.

D

Sol.

$$\vec{E} = -\frac{\partial V}{\partial x} \hat{i} - \frac{\partial V}{\partial y} \hat{j} - \frac{\partial V}{\partial z} \hat{k}$$

$$= -2k(x\hat{i} + y\hat{j} + z\hat{k})$$

$$\vec{a} = \frac{q\vec{E}}{m}$$

$$\text{So, } a = \frac{2k\sqrt{3}q}{m}$$

13.

B

Sol.

$$Q = 6.7 + \frac{(6.7)(4)}{214} = 6.825 \text{ MeV}$$

14.

C

Sol.

It activates due to electron hole pair formation after radiation is incident on depletion region.

15.

B

Sol.

$$I = I_0 \cos^2 \frac{\omega_1 - \omega_2}{2} t$$

$$= I_0 \cos^2 \frac{2\pi(n_1 - n_2)t}{2} = I_0 \cos^2 \pi(4) \frac{1}{16} = \frac{I_0}{2}$$

16.

D

Sol.

$$-\frac{GMm}{2R} \left(3 - \frac{R^2}{4R^2} \right) + \frac{1}{2} m v_e^2 = 0$$

$$-\frac{GM}{2R} \left(\frac{11}{4} \right) + \frac{1}{2} v_e^2 = 0$$

$$v_e = \sqrt{\frac{11GM}{4R}}$$

17. B
Sol. The circuit is in resonance
So rms potential difference across resistance is 220 V
So, $I = \frac{220}{100} = 2.2 \text{ A}$

18. C
Sol. Yellow $\rightarrow 4$
Violet $\rightarrow 7$
Brown $\rightarrow$ multiple 10
Gold $\rightarrow$ Tolerance 5%

19. D
Sol. $\mu mgR = R \frac{\pi R^2}{2\pi R} \frac{d}{dt} (B_0 t^2)$ at $t = 2 \text{ sec}$
So, $\mu = \frac{2B_0 RQ}{mg}$

20. C
Sol. $F_p = 6\pi\eta r(3v_0 - v_0)$ towards the downward direction
So, $P = (12\pi\eta r v_0)(v_0) = 12\pi\eta r v_0^2$

SECTION - B

21. 45
Sol. $I = (80) \left(\frac{1}{2} \right) (\cos 30^\circ)^2 (\cos 30^\circ)^2 = 22.50 \text{ W/m}^2$

22. 25
Sol. from work energy theorem
 $W_E = K_f - K_i$
 $qEa = \frac{1}{2} m [4^2 + 3^2] = \frac{25m}{2}$
 $a = \frac{25m}{2qE} = \frac{25}{2 \times 2 \times 10^{-2} \times 100}$
 $a = 6.25$

23. 15
Sol. $y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$
 $y = x \tan 30^\circ - \frac{(10)x^2}{(2)(10)^2 \frac{3}{4}}$
 $y = \frac{x}{\sqrt{3}} - \frac{2}{30} x^2$
So, $\frac{A}{B} = \frac{15}{\sqrt{3}} = 5\sqrt{3}$

24. 3113

Sol. Total surface area = $6(7.203)^2 = 311.299 = 311.3 \text{ m}^2$

25. 1

Sol. For angular impulse theorem

$$\tau t = I\omega - 0$$

$$\tau = \frac{(2)(60)(2\pi)}{(40) \times (60)} = \frac{\pi}{10} \text{ N-m}$$

$$\Rightarrow \pi\tau = \frac{\pi^2}{10} = 1$$

26. 30

Sol. $(1)(6) = (4 + 1 + 1)v$, at the maximum height

So, $v = 1 \text{ m/s}$

Where v is velocity of the system along the horizontal

For conservation of mechanical energy

$$\frac{1}{2}(2)(3)^2 = \frac{1}{2}(6)(1)^2 + (2)(g)(h)$$

$$h = \frac{9-3}{2g} = 0.3 \text{ m} = 30 \text{ cm}$$

27. 500

Sol. $S = \frac{\sigma 4\pi R_s^2 T_s^4}{4\pi d^2} = \sigma T_s^4 \left(\frac{R_s}{d} \right)^2$

$$2600 = 5.67 \times 10^{-8} T_s^4 \left(\frac{7 \times 10^8}{1.1 \times 10^{11}} \right)^2$$

$$T_s^4 = \frac{2600 \times 10^{22} \times 1.21}{49 \times 10^{16} \times 5.67 \times 10^{-8}} = 0.113 \times 10^{16}$$

$$T_s = 5800 \text{ K}$$

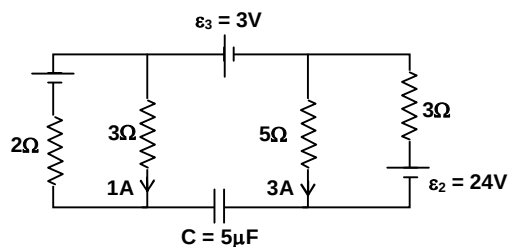
$$\text{So, } \lambda_{\text{mp}} = \frac{2.9 \times 10^{-3}}{5800} = 500 \text{ nm}$$

28. 1125

Sol. In steady state

So, P.D. across the capacitor = 15V

$$\text{So, } U_0 = \frac{1}{2}(5)(15)^2 \mu\text{J} = 562.50 \mu\text{J}$$



29. 1480

Sol. Displaced current = $\frac{i}{2} = \frac{8e^{-\frac{4}{2}}}{2} = 1.48 \text{ A} = 1480 \text{ mA}$

30. 502

Sol. $v = \left(\frac{1500 + 7.5}{1500 + 5} \right) \left(\frac{1500 - 5}{1500 - 7.5} \right) \times 500 = 501.67 \text{ Hz}$

Chemistry

PART – B

SECTION – A

31. B

 Sol. $\text{Cl} \rightarrow \text{strong } (-I), \text{SO}_3^- \rightarrow \text{weak } (-I)$
 $+I : -\text{O}^- > -\text{COO}^- > -\text{CH}_3$

32. A

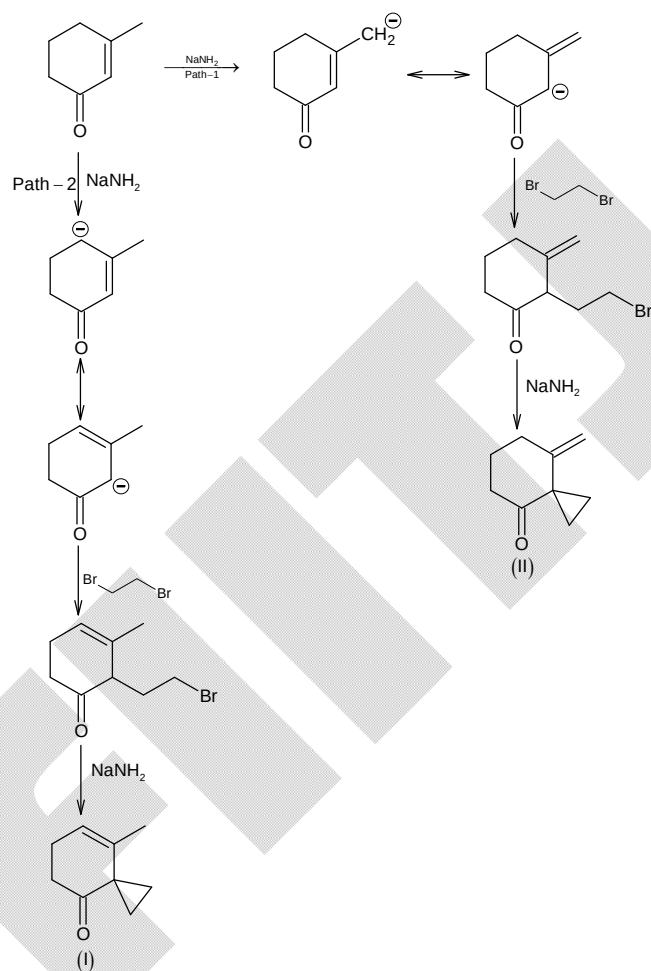
Sol. No chiral centre is present. POS is absent and so the molecule is chiral.

33. D

 Sol. NH_3 is more basic than H_2O . NH_3 ionises as NH_4^+ and NH_2^- .

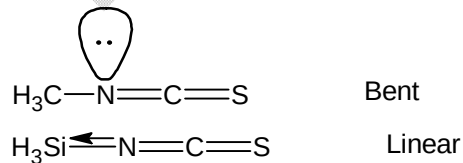
34. C

Sol.

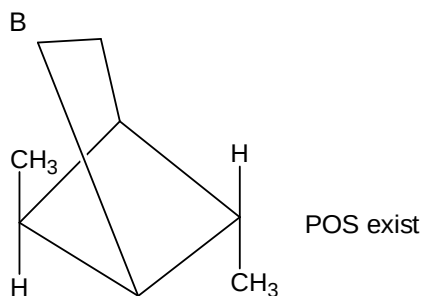


35. D

Sol.



36.
Sol.



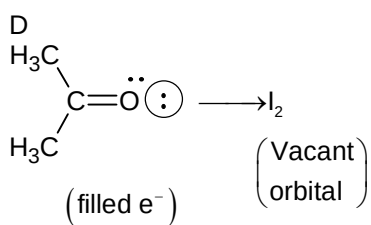
37.
Sol.

C
C – H bond is directly involved in dissociation to form enol for addition of Br⁺

38.
Sol.

A
Factual

39.
Sol.



Charge transfer

I₂ on dissociation converts to I⁽⁻⁾ and then the colour starts fading out.

40.
Sol.

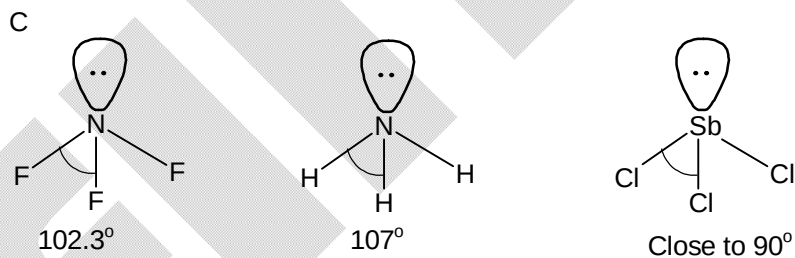
C

$$e^{-\frac{r}{na_0}} = e^{-\frac{r}{3a_0}} \rightarrow \therefore n = 3$$

$$r^{\ell} = r^2 \therefore \ell = 2$$

$\therefore$ 3d orbital.

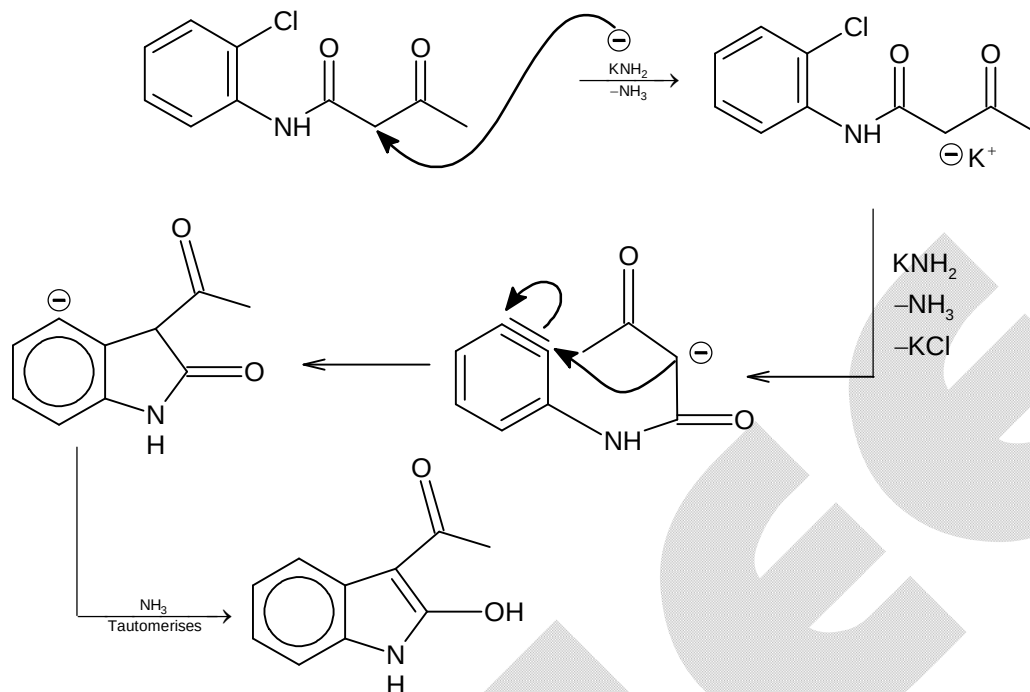
41.
Sol.



42.
Sol.

B
Bond strength of D₂ > H₂.

43. B
Sol.

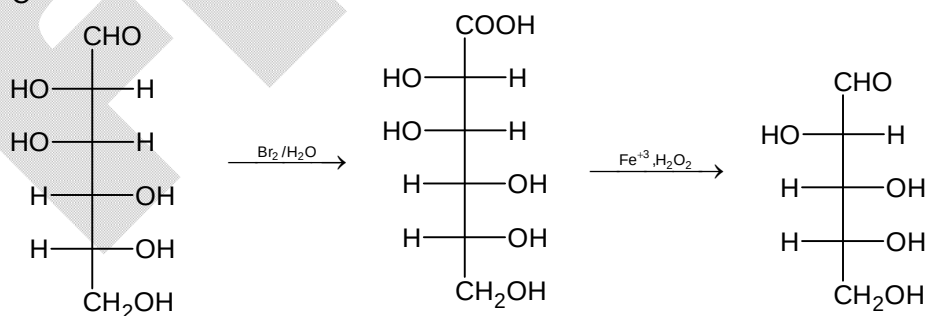


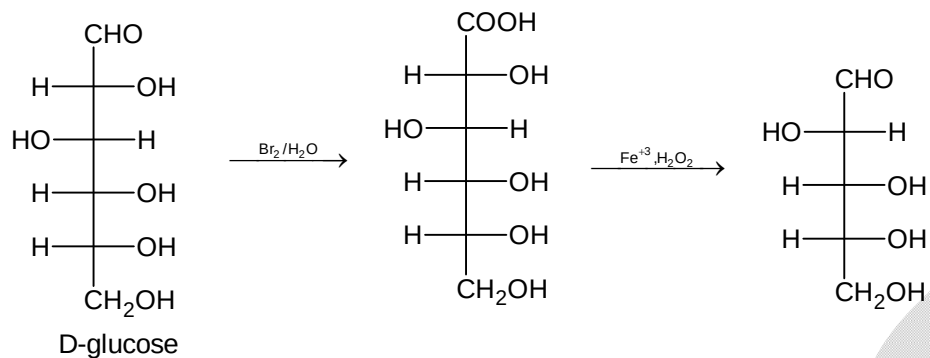
44. B
Sol. Compound 1 : concentration = C_1 ; number of phosphate = C
Compound 2 : concentration = C_2 ; number of phosphate = $2C$.
Total phosphate = 36 m moles in 1 L
 $4C_1 + C_2 = 36$ m moles or $4C + 2C = 36$ m moles
 $C = 6$ m moles
 $\therefore C_1 = [\text{compound 1}] = 24 \text{ mM}$
 $C_2 = [\text{compound 2}] = 6 \text{ mM}$

45. D
Sol. Bouveault-Blanc reduction is used to reduce esters to alcohols.

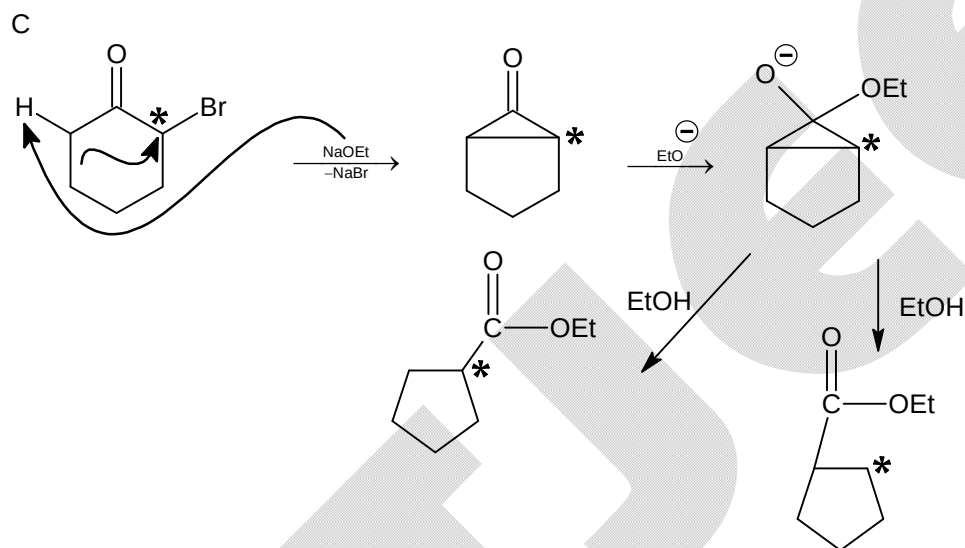
46. B
Sol. The technique is used to separate liquids that are miscible and differ slightly in boiling point.

47. C
Sol.





48.
Sol.



49.
Sol.

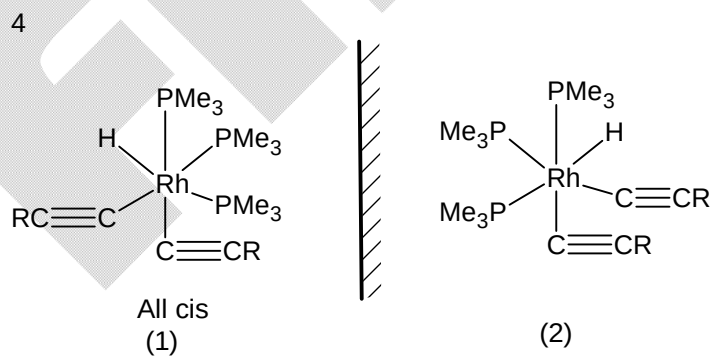
A
Salt bridge allows flow of ions.

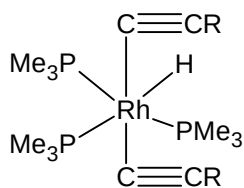
50.
Sol.

B
Solid HF has zig-zag chains of HF molecules.
Intermolecular distance = 155 pm.

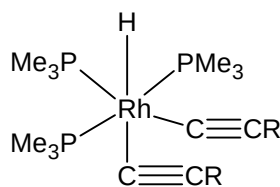
SECTION - B

51.
Sol.



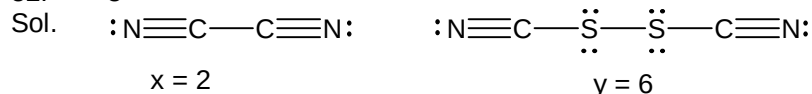


(Inactive)
(3)



(Inactive)
(4)

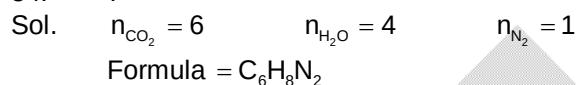
52. 8



53. 168

Sol. $\Delta H^\circ = -nF \left[E^\circ - T \left(\frac{dE^\circ}{dT} \right) \right]$
 $= -2 \times 96500 \left[0.675 - 300 \times \left[-6.5 \times 10^{-4} \right] \right]$
 $= -167.910 \text{ kJ}$

54. 7



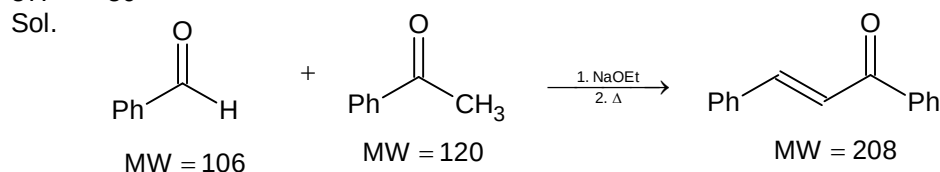
55. 6

Sol. $\text{pK}_{a_1} = 2.40$ $\text{pK}_{a_2} = 9.60$
 $\text{pH} = \frac{1}{2} (\text{pK}_{a_1} + \text{pK}_{a_2})$

56. 28

Sol. $\Delta U_{ABC} = \Delta U_{AB} + \Delta U_{BC} + (-\Delta U_{AC})$
 $\Delta U_{AC} = \Delta U_{AB} + \Delta U_{BC}$
 Now, $\Delta U_{AB} = q_{AB} + w_{AB} = 600 \text{ J} + 0$
 $\Delta U_{BC} = q_{BC} + w_{BC} = 200 + (-P\Delta V)$
 $= 200 + \left[-8 \times 10^4 \frac{\text{N}}{\text{m}^2} (3 \times 10^{-3} \text{ m}^3) \right]$
 $= 200 + [-240 \text{ J}]$
 $= -40 \text{ J}$
 $\therefore \Delta U_{AC} = 600 - 40 = 560 \text{ J} = x$
 $\therefore \frac{x}{20} = 28$

57. 50



$$n_{\text{PhCHO}} = \frac{10.6}{106} = 0.1$$

1 mole of PhCHO $\longrightarrow$ 1 mole product

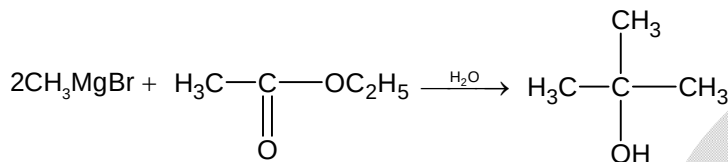
$$n_{\text{Product(produced)}} = 0.1 = \frac{W_{\text{expected}}}{208}$$

$$W_{\text{expected}} = 20.8 \text{ g}$$

$$\% \text{yield} = \left(\frac{10.4}{20.8} \right) \times 100 = 50\%$$

58. 25

Sol.



$$10 \text{ mL} \times 0.025 = 0.25 \text{ mmole} = n_{\text{CH}_3\text{COC}_2\text{H}_5}$$

$$n_{\text{MeMgBr}} = 0.5 \text{ mmole} = M \times V$$

$$V = \frac{0.5 \text{ mmole}}{0.2 \text{ M}} = 2.5 \text{ mL}$$

$$x = 2.5 \text{ mL}$$

$$10x = 25$$

59. 4

Sol.

$$\frac{dP}{dt} = \frac{k_1 k_2 [E_0][S]}{k_1 [S]} = k_2 [E_0] = 0.02$$

$$\rightarrow \frac{k_2 [E_0]}{2} = \frac{k_1 k_2 [E_0][S]}{k_{-1} + k_1 [S]}$$

$$\Rightarrow k_{-1} + k_1 [S] = 2k_1 [S]$$

$$\Rightarrow k_{-1} = k_1 [S]$$

$$\frac{k_1}{k_{-1}} = \frac{1}{[S]} = \frac{1}{(250 \times 10^{-6} \text{ kg/dm}^3)} = 4000$$

60. 5

Sol.

Let the concentration of FeSO_4 and K_2S be $x\text{M}$ and the volume is $V \text{ L}$



$$\begin{array}{ccccccc} x \times V & x \times V & - & - & & & \\ - & - & x \times V & x \times V & & & \end{array}$$

$$K_{\text{sp}} = [\text{Fe}^{+2}][\text{S}^{-2}]$$

$$6.3 \times 10^{-18} = \left(\frac{x \times V}{2V} \right)^2$$

$$x = 5.02 \times 10^{-9}$$

$$\therefore C = 5.02$$

Mathematics

PART – C

SECTION – A

61. B

Sol. $\lim_{|x| \rightarrow \infty} \frac{2x^2 - kx + 5}{x^2 + 3x + 4} = 2$
 so, there must be some k s.t.
 $\frac{2x^2 - kx + 5}{x^2 + 3x + 4} = 2 \Rightarrow k \neq -6$

62. A

Sol. Let $\alpha < \beta$, $f(x) = \cot^{-1}x$
 Using L.M.V.T $\frac{\cot^{-1}\alpha - \cot^{-1}\beta}{\alpha - \beta} = -\frac{1}{1+c^2} \Rightarrow \frac{\alpha - \beta}{\cot^{-1}\alpha - \cot^{-1}\beta} = -(c^2 + 1)$
 $\sqrt{2} < \alpha < c < \beta < \sqrt{3} \Rightarrow 2 < c^2 < 3$
 $\Rightarrow 3 < c^2 + 1 < 4$
 $-4 < -(c^2 + 1) < -3$

63. C

Sol. As $\{a\} + \left\{\frac{1}{a}\right\} = 1 \Rightarrow \{a^3\} + \left\{\frac{1}{a^3}\right\} = 1$

So, $\int_0^1 \frac{x}{1+x} \sqrt{1-x^2} dx$

Let $x = \sin\theta$, $dx = \cos\theta.d\theta$

$\int_0^{\pi/2} \frac{\sin\theta \cdot \cos^2\theta d\theta}{1+\sin\theta} = \int_0^{\pi/2} \frac{\sin\theta(1-\sin^2\theta)}{(1+\sin\theta)} d\theta$

$\int_0^{\pi/2} \sin\theta(1-\sin\theta) d\theta = 1 - \frac{\pi}{4}$

64. B

Sol. $a(\sin x + 2\sin x \cdot \cos x) = 3\sin x - 4\sin^3 x$
 $= \sin x(3 - 4\sin^2 x)$
 $= \sin x(3 - 4(1 - \cos^2 x))$
 $a(\sin x + 2\sin x \cdot \cos x) = \sin x(4\cos^2 x - 1)$
 As $\sin x \neq 0$, in $(0, \pi)$
 $\Rightarrow a(1 + 2\cos x) = 4\cos^2 x - 1$
 $\Rightarrow 4\cos^2 x - 2a\cos x - (1+a) = 0$
 $\Rightarrow \cos x = \frac{a \pm (a+2)}{4} \Rightarrow \cos x = -\frac{1}{2} \text{ and } \frac{a+1}{2}$
 Since $\cos x = -\frac{1}{2}$
 $x = \frac{2\pi}{3}$
 for $\cos x = \frac{a+1}{2} \Rightarrow -1 < \frac{a+1}{2} < 1 \quad a \in (-3, 1)$

if $\frac{a+1}{2} = -\frac{1}{2} \Rightarrow a = -2$

this will repeat one solution so $a \in (-3, 1) - \{-2\}$

65.

B

Sol.

$$z^2 - cz + 10 = 0$$

and z_1, z_2 are roots and c is real so $z_2 = \bar{z}_1$

$$\text{We have } z_1 z_2 = 10, \Rightarrow z_1 \bar{z}_1 = 10 \Rightarrow |z_1|^2 = 10$$

$$\text{So, } |z_1| = \sqrt{10}$$

$$\text{and } z_1 + z_2 = c$$

$$z_1 + \bar{z}_1 = c$$

$$2\text{Re}(z_1) = c \Rightarrow \text{Re}(z_1) = \frac{c}{2}, \text{Im}(z_1) = \sqrt{10 - \frac{c^2}{4}}$$

$$\text{Now } \frac{1}{z_1} = \frac{1}{10} \cdot \frac{10}{z_1} = \frac{1}{10} \cdot \frac{z_1 z_2}{z_1} = \frac{z_2}{10} = \frac{\bar{z}_1}{10}$$

$$\frac{1}{z_2} = \frac{z_1}{10}$$

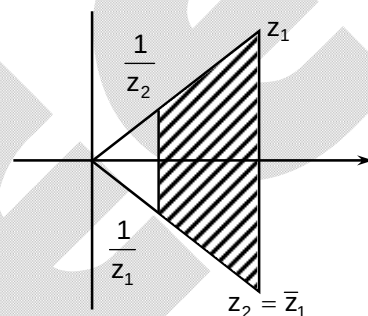
$$\text{Now area of quadrilateral} = \frac{1}{2} |\text{Re}(z_1)|^2 |z_1| \left(1 - \frac{1}{10^2}\right)$$

$$= \frac{1}{2} |c| \sqrt{10 - \frac{c^2}{4}} \left(1 - \frac{1}{10^2}\right)$$

$$\Rightarrow \frac{1 - \frac{1}{10^2}}{2} \sqrt{\left(\frac{c^2(40 - c^2)}{4}\right)} \text{ is maximum}$$

$$\text{if } c^2 = 40 - c^2$$

$$c^2 = 20 \Rightarrow c = 2\sqrt{5}$$



66.

C

Sol.

$$\frac{\alpha^2}{1 - \tan^2 x} = \frac{(\sin^2 x + \alpha^2 - 2)(1 + \tan^2 x)}{1 - \tan^2 x}$$

$$\Rightarrow \alpha^2 \cos^2 x = \sin^2 x + \alpha^2 - 2$$

$$\Rightarrow \sin^2 x = \frac{2}{1 + \alpha^2} \Rightarrow 0 \leq \frac{2}{1 + \alpha^2} \leq 1 \Rightarrow \alpha^2 \geq 1$$

67.

A

Sol.

Let polynomial is $P(x)$ and two integer roots are α, β

$\Rightarrow P(x) = (x - \alpha)(x - \beta)(x^2 + ax + b)$ for some integer a, b if $p + iq$ is a root of $P(x)$, then $p - iq$ is also a root (with $q \neq 0$).

$$\Rightarrow x^2 + ax + b = (x - (p + iq))(x - (p - iq)) = x^2 - 2px + p^2 + q^2$$

Product of the roots is $p^2 + q^2$ sum $2p$

$$a = -2p, b = p^2 + q^2$$

$$\text{from option } \frac{1+i\sqrt{11}}{2}, \frac{1-i\sqrt{11}}{2} \text{ is only roots.}$$

68.

C

Sol.

Given circle $x^2 + y^2 - 6x - 4y + 11 = 0$ (centre $(3, 2)$ $r = \sqrt{2}$ tangent at $(2, 3)$ will

$$2x + 3y - 3(x + 2) - 2(y + 3) + 11 = 0$$

$$-x + y - 1 = 0 \Rightarrow x - y + 1 = 0$$

$$\text{Coordinate of } A \left(3 + 4 \frac{1}{\sqrt{2}}, 2 + 4 \frac{1}{\sqrt{2}} \right) = (3 + 2\sqrt{2}, 2 + 2\sqrt{2})$$

Reflection of A with respect to line is

$$\frac{x - (3 + 2\sqrt{2})}{1} = \frac{y - (2 + 2\sqrt{2})}{-1} = -2 \left(\frac{2}{2} \right) = -2$$

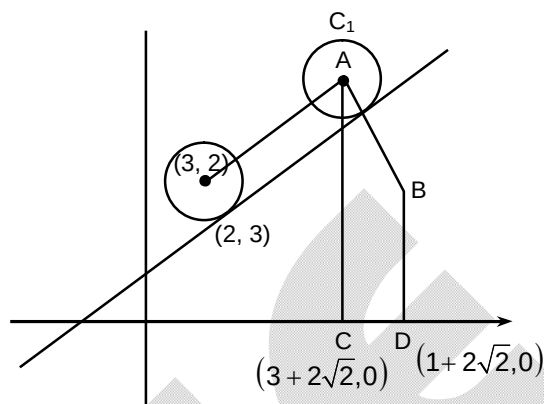
$$x = -2 + 3 + 2\sqrt{2} = 1 + 2\sqrt{2}$$

$$y = 4 + 2\sqrt{2}$$

$$B(1 + 2\sqrt{2}, 4 + 2\sqrt{2})$$

Area of ABCD

$$= \frac{1}{2} (2 + 2\sqrt{2} + 4 + 2\sqrt{2}) 2 = (6 + 4\sqrt{2})$$



69.

D

Sol.

The angle APB is obtuse if and only P lies inside the circle with diameter AB

$$\text{The area of } \triangle AFB \text{ is } = \frac{1}{2} \times 2 \times 4 = 4$$

$$\text{Area of ABCDE} = 4(2\pi + 1) - 4 = 8\pi$$

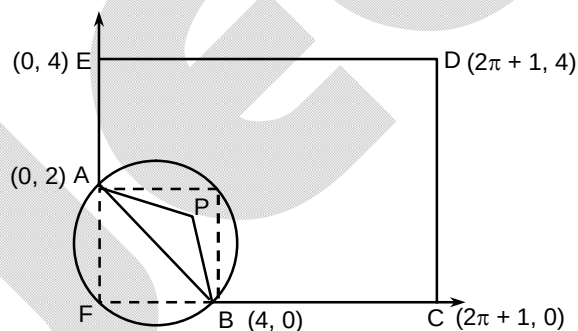
$$AB = 2\sqrt{5}$$

$$\text{Radius of circle} = \sqrt{5}$$

$$\text{Area of circle inside ABCDE is } \frac{5\pi}{2}$$

Probability is that $\angle APB$ is obtuse

$$= \frac{5\pi}{2 \times 8\pi} = \frac{5}{16}$$



70.

A

Sol.

B's . C's	A's . C's	A's . B's
1 2 3 4 5	6 7 8 9 10	11 12 13 14 15
k B's (5 - k) C's	(5 - k) A's k C's	k' A's (5 - k) B's

If there are K B's in the 1st five letters, then there must be (5 - k) is in the first five letter so there must be K' C's and (5 - K) A's in the next five letters and K A's and (5 - K) B's in the last five letter.

So the arrangement is

$$\sum_{k=0}^5 \left(\frac{5!}{k!(5-k)!} \right) \times \frac{5!}{k!(5-k)!} \times \frac{5!}{k!(5-k)!}$$

71.

C

Sol.

$$\begin{aligned} & \sum_{i=0}^n \sum_{j=1}^n {}^nC_j \cdot {}^jC_i \\ &= {}^nC_1 ({}^1C_0 + {}^1C_1) + {}^nC_2 ({}^2C_0 + {}^2C_1 + {}^2C_2) + {}^nC_3 ({}^3C_0 + {}^3C_1 + {}^3C_2 + {}^3C_3) \dots + {}^nC_n ({}^nC_0 + {}^nC_1 + \dots + {}^nC_n) \\ &= 2^n {}^nC_1 + 2^2 \cdot {}^nC_2 + 2^3 \cdot {}^nC_3 + \dots + 2^n \cdot {}^nC_n \\ &= (1 + 2)^n - 1 = 3^n - 1 \end{aligned}$$

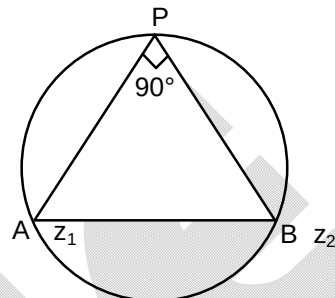
72. C

Sol. Let $P(x) = ax^2 + bx + c$ reflect it over the x-axis we have $-ax^2 - bx - c$
 so $f(x) = a(x+5)^2 - b(x-5) - c = -ax^2 + 10ax - bx - 25a + 5b - c$
 $f(x) + g(x) = 20ax + 10b$ since $a \neq 0$

73. B

Sol. So locus of P is a circle with diameter $AB = \sqrt{2}$

$$\text{Radius} = \frac{1}{\sqrt{2}}$$



74. B

Sol. $I = \int_{1/10}^{9/10} \left(2x^3 - 3x^2 - x + \frac{3}{2} \right) dx$
 $I = \int_{1/10}^{9/10} \left(2(1-x)^3 - 3(1-x)^2 - (1-x) + \frac{3}{2} \right) dx$
 $\Rightarrow 2I = \int_{1/10}^{9/10} 1 dx = \frac{4}{5} \Rightarrow I = \frac{2}{5}$

75. A

Sol. Let $1 - x^3 = t$

$$\Rightarrow 3I_1 = \int_0^1 (1-t^{\sqrt{2}})^{\sqrt{3}} dt \text{ and } 3I_2 = \int_0^1 (1-t^{\sqrt{2}})^{\sqrt{3}+1} dt$$

Applying integration by parts on $3I_2$

$$3I_2 = (1-t^{\sqrt{2}})^{\sqrt{3}+1} \times t \Big|_0^1 - (\sqrt{3}+1) \int_0^1 (1-t^{\sqrt{2}})^{\sqrt{3}} \times -\sqrt{2} t^{\sqrt{2}-1} \times t dt$$

$$\Rightarrow 3I_2 = -(\sqrt{3}+1)\sqrt{2} \int_0^1 (1-t^{\sqrt{2}})^{\sqrt{3}} (1-t^{\sqrt{2}}-1) dt$$

$$\Rightarrow 3I_2 = -(\sqrt{3}+1)\sqrt{2} \times 3I_2 + (\sqrt{3}+1)\sqrt{2} \times 3I_1$$

$$\Rightarrow I_2 = (1 + (\sqrt{3}+1)\sqrt{2})(\sqrt{3}+1)\sqrt{2} I_1$$

$$\Rightarrow \frac{I_1}{I_2} = \frac{1 + (\sqrt{3}+1)\sqrt{2}}{(\sqrt{3}+1)\sqrt{2}} = 1 + \frac{1}{(\sqrt{3}+1)\sqrt{2}}$$

76. B

Sol. Let $f(x) = \int_0^x \frac{t^{10} + 1}{t^{10} + t^4 + 1} dt = -6x + 5$

$f(0) = 5$ and $f(1)$ is negative

$$f'(x) = \frac{x^{10} + 1}{x^{10} + x^4 + 1} - 6 = -5 - \frac{x^4}{x^{10} + x^4 + 1}$$

$\therefore f'(x)$ is always negative $\therefore$ only 1 solution.

77. A

$$\text{Sol. } f(x+y) = \frac{f(x)f(y)}{f(x)+f(y)}$$

$$\Rightarrow f(x+y) = \frac{1}{\frac{1}{f(x)} + \frac{1}{f(y)}}$$

$$\Rightarrow \frac{1}{f(x+y)} = \frac{1}{f(x)} + \frac{1}{f(y)}$$

$$\text{Let } \frac{1}{f(x)} = g(x)$$

$$\Rightarrow g(x+y) = g(x) + g(y) \quad \forall x, y \in \mathbb{R}$$

$$\Rightarrow g(x) = kx, \text{ where } k \text{ is constant}$$

$$\Rightarrow \frac{1}{f(x)} = kx$$

$$\Rightarrow f(x) = \frac{1}{kx} \text{ but } f(1) = 32 \Rightarrow k =$$

$$\Rightarrow f(x) = \frac{32}{x}$$

$$f(4) = \frac{32}{4} = 8$$

78. D

$$\text{Sol. This can be written as } \lim_{x \rightarrow \infty} \left(\frac{\sin \frac{5}{|1-a^4|^x}}{\frac{5}{|1-a^4|^x}} \right)^5$$

$$\text{since } a^2 > 2$$

$$a^4 > 4$$

$$-a^4 < -4$$

$$1 - a^4 < -3$$

$$|1 - a^4| > 3$$

79. D

$$\text{Sol. } g(x) = 2f\left(\frac{x}{2}\right) + f(2-x)$$

$$\Rightarrow g'(x) = 2f'\left(\frac{x}{2}\right) \cdot \frac{1}{2} + f'(2-x)(-1)$$

$$\Rightarrow g'(x) = f'\left(\frac{x}{2}\right) - f'(2-x)$$

$$\text{Given } f''(x) < 0 \quad \forall x \in (0, 2) \Rightarrow f'(x) \text{ is decreasing in } (0, 2)$$

$$\text{Case I : If } \frac{x}{2} > 2-x \text{ and } f'(x) \text{ is decreasing}$$

$$f'\left(\frac{x}{2}\right) < f'(2-x) \quad \forall x > \frac{4}{3}$$

$$\Rightarrow g'(x) \text{ is decreasing in } \left(\frac{4}{3}, 2\right)$$

Case II : If $\frac{x}{2} < 2 - x$ and $f'(x)$ is decreasing

$$\Rightarrow f'\left(\frac{x}{2}\right) > f'(2-x) \quad \forall x \in \left(0, \frac{4}{3}\right)$$

$g(x)$ increases in $\left(0, \frac{4}{3}\right)$.

80. B

Sol. Area bounded = $\int_0^{\pi/4} (1 - \cos x) dx + \int_{\pi/4}^{5\pi/4} (1 - \sin x) dx + \int_{5\pi/4}^{2\pi} (1 - \cos x) dx = 2\pi - 2\sqrt{2}$

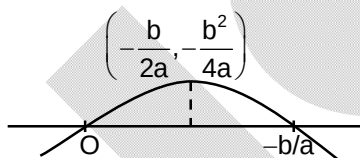
SECTION - B

81. 2

Sol. $f(x) = \sqrt{x(ax+b)}$ co-domain is $[0, \infty)$

Domain also $x \geq 0$

$$f(x) = 0, x = 0, -\frac{b}{a}$$



$$\text{so } -\frac{b}{a} > 0 \Rightarrow a < 0$$

$$-\frac{b}{a} = \sqrt{-\frac{b^2}{4a}} \Rightarrow -\frac{b}{a} = \frac{b}{\sqrt{-4a}}$$

$$-a = \sqrt{-4a} \Rightarrow a^2 = -4a$$

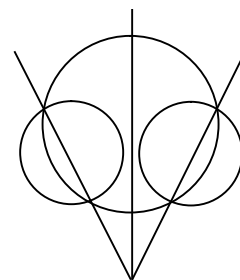
$$a^2 + 4a = 0$$

$$a = 0, -4$$

82. 2

Sol. So radical axis will be concurrent $\begin{vmatrix} 2 & 2 & 3 \\ 1 & 1 & 4 \\ 2 & k & 2 \end{vmatrix} = 0$

$$\Rightarrow k = 2$$



83. 5

Sol. Possible values are $\{-2, -1, 0, 1, 2\}$

84. 2

Sol. Since, sum of all numbers is $1 + 2 + 3 + \dots + 9 = 45$, each groups sum will be $= \frac{45}{3} = 15$
groups are (9, 1, 5) (8, 3, 4) (7, 2, 6) and (9, 2, 4) (8, 1, 6) (7, 3, 5).

85. 8

Sol. Let the common diff. between consecutive a_i be 'd'

$$\text{then } a_{15} = a_1 + 14d$$

$$a_{15} - a_1 = 14d$$

$$\text{Since } 1 \leq a_1 \leq 10, 241 \leq a_{15} \leq 250$$

$$231 \leq a_{15} - a_1 \leq 249 \Rightarrow 231 \leq 14d \leq 249$$

$$\text{and } d \text{ must be an integer } \Rightarrow d = 17$$

$$\text{Now, } 1 \leq a_1 \leq 10, 13 \leq a_2 \leq 20, 241 \leq a_1 + 238 \leq 250$$

$$1 \leq a_1 \leq 10, 13 \leq a_1 + 17 \leq 20, -4 \leq a_1 \leq 3, 3 \leq a_1 \leq 12$$

$$\Rightarrow a_1 = 3$$

$$\text{So, } a_{14} = 3 + 13 \times 17 = 224.$$

86. 5

Sol. Let the first 2 term is x and y, then the terms would be x, y, xy, xy^2 , x^2y^3 and x^3y^5

$$\Rightarrow x^3y^5 = 4000 = 5^3 \cdot 2^5$$

$$\Rightarrow x = 5, y = 2.$$

87. 36

Sol. $1 + 7 + 5 + 2 + 5 = 20$

So, there are 3 possible cases :

$$(1) \text{ if mean is 5 } \Rightarrow x = 5 \times 6 - 20 = 10$$

$$(2) \text{ if mean is 7 } \Rightarrow x = 7 \times 6 - 20 = 22$$

$$(3) \text{ if mean is } x \Rightarrow x = \frac{20+x}{6} \Rightarrow x = 4$$

$$\text{So, answer} = 10 + 22 + 4 = 36$$

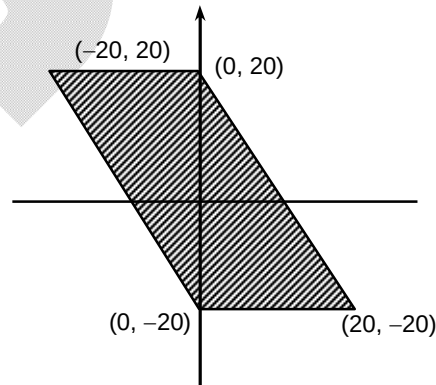
88. 800

$$\text{Sol. } y^2 + 2xy + x^2 = 400 - 40|x| + x^2$$

$$(x+y)^2 = (20-|x|)^2$$

$$x+y = |x| - 20$$

$$\text{or } x+y = 20 - |x|$$



89. 841

Sol. All possible location of P are integral points such that $|x| + |y| \leq 20$

$$\text{i.e. } 1 + 3 + 5 + 7 + \dots + 41 + 39 + 37 + \dots + 1$$

$$= 20^2 + 21^2 = 29^2 = 841$$

90. 61

Sol. The current range = $28 - 3 = 25$

To double the range that is = 50

Since we do not want to change the median, we need to get a value less than or equal to 8 for the smaller, making 53 fixed for the larger, since anything less than 3 is not beneficial to the optimization. So taking our optimal values of 8 and 53, we have an answer $8 + 53 = 61$.